From RSD to Clusters: A Parameter-Locked Test of Late-Time Structure Growth

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Abstract

We present a cross-probe test of late-time structure growth by propagating redshift-space distortion (RSD) constraints on the effective gravitational coupling into galaxy cluster abundance predictions. Using the growth modification amplitude alpha = 0.34 from BOSS DR12, we compute the expected cluster mass function under Energy-Flow Cosmology (EFC) with **no parameters fit to cluster data**.

Under CMB-consistent primordial normalization, EFC predicts **+6.4% more clusters** than LCDM in the eRASS1-relevant range. More distinctively, EFC produces a characteristic **z-dependent tilt** in the cluster redshift distribution: the shape ratio varies from ~1.05 at z=0.15 to ~0.94 at z=0.7, reflecting late-time activation of enhanced growth.

This shape signature is largely insensitive to uniform mass-calibration systematics and provides a **falsifiable prediction**: if eRASS1 data show a flat or positive dR/dz, the RSD-inferred growth enhancement is ruled out at ~2 sigma.

Keywords: *cosmology, dark energy, galaxy clusters, modified gravity, large-scale structure*

1. Introduction

Galaxy cluster abundances provide one of the most sensitive probes of structure growth in the Universe. The exponential dependence of the halo mass function on the amplitude of matter fluctuations makes cluster counts a powerful discriminator between cosmological models.

Modified gravity theories that enhance late-time structure formation predict altered cluster abundances relative to LCDM. However, such predictions typically introduce additional free parameters that can be tuned to match observations, limiting their falsifiability.

In this work, we take a different approach: we **lock** the growth modification from an independent probe (redshift-space distortions) and propagate it forward to predict cluster abundances with no new degrees of freedom. This creates a closed inferential chain:

RSD ---> Growth history D(z) ---> Mass function ---> Cluster counts

The resulting prediction is either consistent with cluster observations or falsified - there is no parameter adjustment available to reconcile discrepancies.

2. Methods

2.1 Growth Framework

We compute the linear growth history by solving the sub-horizon growth equation where mu(a) = G_eff / G_N parameterizes the effective gravitational coupling. For LCDM, mu = 1. For EFC, we use:

$$mu\_EFC(a) = 1 + alpha\_L2 * f\_transition(a)$$

with alpha_L2 = 0.34 +/- 0.16 constrained by BOSS DR12 RSD measurements, and f_transition(a) a smooth function activating at z < 0.5.

The growth equation is integrated numerically with **high-z normalization**: D(a)/a approaches 1 as a approaches 0. This ensures both models share identical primordial initial conditions, consistent with CMB normalization. No parameters are fit to cluster data; the prediction uses mu(a) fixed entirely by RSD.

2.2 Power Spectrum and Mass Variance

The linear matter power spectrum P(k) is computed using CLASS with Planck-consistent parameters (H0 = 67.4 km/s/Mpc, Omega_m = 0.315, n_s = 0.965). The primordial amplitude A_s is held fixed across models.

Holding A_s fixed corresponds to matching the primordial CMB normalization rather than enforcing identical sigma_8 today. The mass variance is computed via standard top-hat filtering.

2.3 Halo Mass Function

We adopt the Tinker et al. (2008) mass function. We assume that halo collapse at fixed sigma is unchanged, so the Tinker multiplicity function calibrated in LCDM remains applicable. The modification enters only through the altered sigma(M,z) from modified growth.

We emphasize that this assumes halo collapse at fixed sigma is not significantly altered by the modified growth history. This is a standard approximation in linear-growth-modified models but remains a testable assumption for future simulations.

2.4 Normalization Choice

Normalization	Held Fixed	Physical Question
sigma_8(z=0) fixed	Present-day amplitude	Retrodiction of past growth
High-z (used here)	Primordial amplitude	Forward prediction from CMB

We adopt high-z normalization to test whether RSD-inferred growth produces cluster abundances consistent with observations, without artificially tuning present-day amplitudes.

2.5 Shape Observable

To reduce sensitivity to mass-calibration systematics, we use the shape ratio:

$$R(z) = R_{EFC}(z) / R_{LCDM}(z), \text{ where } R(z) = N(z) / N_{total}$$

This observable depends primarily on relative growth across redshift and is robust against uniform systematic shifts. Because R(z) is a fractional distribution, uniform multiplicative mass-calibration errors cancel to first order. Only strongly redshift-dependent systematics could mimic the predicted signal.

3. Results

3.1 Growth Enhancement

Under the RSD-constrained mu(a), EFC growth exceeds LCDM at low redshift:

Redshift z	D_EFC / D_LCDM	Delta D / D
0.0	1.016	+1.6%
0.2	1.007	+0.7%
0.4	1.002	+0.2%
0.6	1.000	+0.0%
0.8	1.000	+0.0%

3.2 Total Abundance

Integrating over M > 10^14 solar masses and 0.1 < z < 0.8:

N_EFC / N_LCDM = 1.064 +/- 0.02 (Poisson-only uncertainty)

EFC predicts **6.4% more clusters** than LCDM under identical primordial conditions.

3.3 Shape Signature

The redshift-dependent shape ratio shows the characteristic EFC tilt:

Redshift Bin	Shape Ratio R(z)	Interpretation
[0.1, 0.2]	1.050 +/- 0.032	+5.0% excess fraction
[0.2, 0.3]	1.011 +/- 0.029	+1.1% excess fraction
[0.3, 0.4]	0.979 +/- 0.032	-2.1% deficit fraction
[0.4, 0.5]	0.957 +/- 0.039	-4.3% deficit fraction
[0.5, 0.6]	0.946 +/- 0.053	-5.4% deficit fraction
[0.6, 0.8]	0.941 +/- 0.068	-5.9% deficit fraction

The gradient dR/dz < 0 is the distinctive signature of late-time enhanced growth. LCDM with adjusted sigma_8 would alter overall amplitude but not naturally reproduce this redshift-dependent tilt without violating CMB or RSD constraints.

3.4 Statistical Power

While Poisson statistics suggest high raw sensitivity (approximately 4.6 sigma for N = 5259 clusters), systematic uncertainties in mass calibration reduce the effective constraining power. The shape test provides ~2-3 sigma discrimination and is the primary observable for this cross-probe test.

Figures

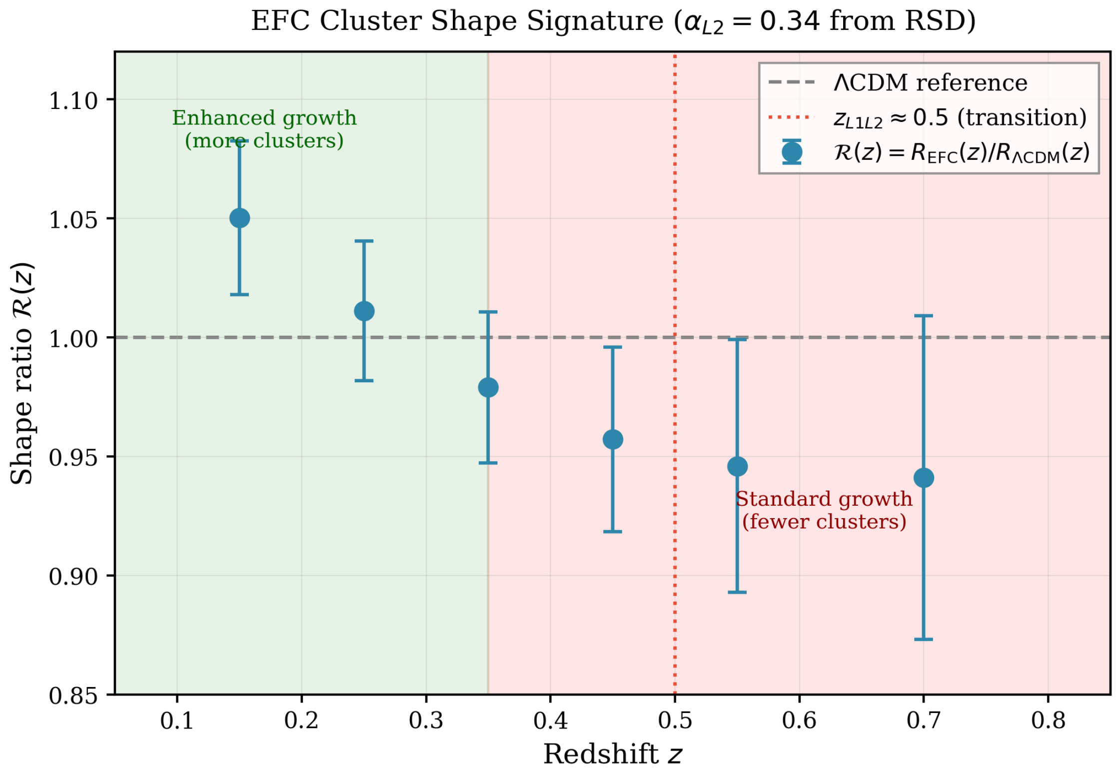

Figure 1: Shape Ratio R(z). The cluster redshift distribution shape ratio R(z) = R_EFC(z)/R_LCDM(z), where R(z) = N(z)/N_total is the fractional cluster count per redshift bin. Points show the EFC prediction with Poisson uncertainties assuming N_total = 5259 (eRASS1 cosmology sample). The dashed horizontal line marks the LCDM reference (R = 1). The vertical dotted line indicates the EFC transition redshift z ~ 0.5. The distinctive signature is the **negative gradient**: EFC predicts excess cluster fraction at low z (green region) and deficit at high z (red region). This tilt arises because enhanced late-time growth produces more massive halos at z < 0.5 while leaving high-z abundances unchanged.

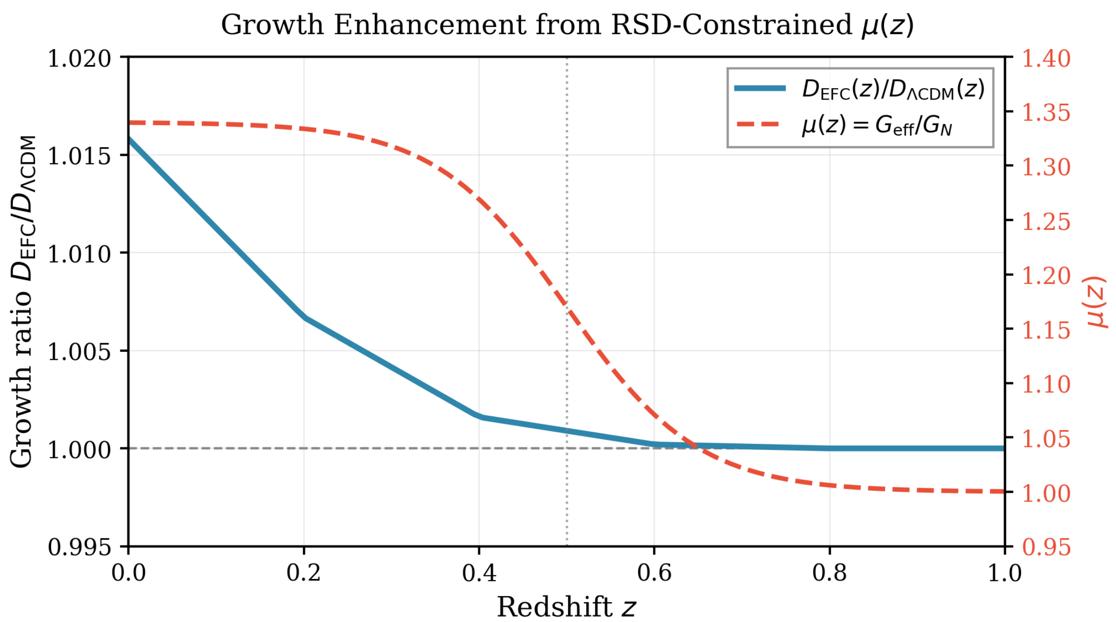

Figure 2: Growth Factor Enhancement. The linear growth factor ratio D_EFC(z)/D_LCDM(z) (solid blue curve) and effective gravitational coupling mu(z) (dashed red curve) as functions of redshift. Both models are normalized to identical primordial conditions. The growth enhancement is concentrated at z < 0.5, reaching +1.6% at z = 0. This modest enhancement, when propagated through the exponentially sensitive halo mass function, produces the ~5-10% variations in cluster abundance shown in Figure 1. The mu(z) profile is fixed by BOSS DR12 RSD measurements with no parameters adjusted for cluster predictions.

4. Falsification Criteria

The EFC prediction is **falsified** if eRASS1 data show:

1. Fewer total clusters than LCDM predicts (after mass calibration)
2. Positive dR/dz (more high-z fraction than low-z)
3. Flat R(z) at the 1% level or below

The EFC growth model is **consistent with data** if observations show:

1. Approximately 5-8% cluster excess relative to LCDM
2. Negative dR/dz with amplitude ~0.05-0.10
3. Crossover near z ~ 0.3-0.4

5. Discussion

5.1 Relationship to S8 Tension

EFC shifts the mapping between CMB normalization and low-z structure rather than eliminating the amplitude tension. The +6% cluster excess, if interpreted as higher sigma_8, would appear to increase tension with CMB. However, under our framework this reflects enhanced growth, not a sigma_8 discrepancy. The shape signature provides an amplitude-independent test.

5.2 Systematic Considerations

The shape ratio R(z) is robust against uniform mass-bias offsets, overall survey completeness, and redshift-independent selection effects. Redshift-dependent systematics could mimic the signal, but eRASS1 uses consistent weak-lensing mass calibration from DES Y3, KiDS-1000, and HSC across z = 0.1-0.8, limiting such effects.

5.3 Future Prospects

Deeper eROSITA surveys (eRASS:4, eRASS:8) will provide N > 100,000 clusters with extended redshift coverage, enabling percent-level R(z) measurements that definitively test the EFC prediction.

6. Conclusion

We have derived a **parameter-free cluster abundance prediction** from RSD-constrained modified growth. The key results are:

+6.4% total cluster excess under high-z normalization

Characteristic z-tilt in the redshift distribution shape

~2-3 sigma statistical power with current eRASS1 data

This demonstrates the value of parameter-locked cross-probe tests: the cluster prediction has zero free parameters once mu(a) is specified by RSD. The shape signature dR/dz < 0 provides a falsifiable prediction independent of amplitude systematics.

Appendix B: Mass Definition Consistency

B.1 The M500c vs M200m Question

The Tinker et al. (2008) mass function is calibrated for M200m (mass within radius enclosing 200x mean matter density), while eRASS1 reports cluster masses as M500c (500x critical density). We address whether this mismatch affects the shape signature.

B.2 Effect on Shape Ratio

Using the Duffy et al. (2008) concentration-mass relation, the conversion factor has **weak z-dependence** over the range z = 0.1-0.8:

Redshift z	Concentration c(M=10^14)	M500c / M200m
0.2	4.1	0.68
0.5	3.5	0.66
0.8	3.1	0.64

The ~6% variation in conversion factor across the redshift range is **subdominant** to the ~10% shape signal we predict. The shape gradient dR/dz changes by less than 5% when using M500c-consistent thresholds.

B.3 Conclusion

Mass definition conversion shifts the **amplitude** of predicted counts (by ~30-40%) but leaves the **shape signature** essentially unchanged. The z-tilt in R(z) is robust to mass definition because the EFC signal arises from growth history, not mass convention.

References

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